

.NET & Razor Syntax

Projede Kullanılan Syntax Ornekleri

1. .NET Attribute'lari
2. async / await
3. Action Donus Tipleri
4. Razor @model & @{ }
5. Razor Kontrol Yapilari
6. Tag Helpers (asp-*)
7. @Html Helper'lari

Koku. - Web Programlama 2. Proje - Ek Dokuman

.NET Syntax Ornekleri

1. Attribute'lar (Controllers/*.cs)

Controller'larda HTTP metod, yetkilendirme ve validasyon icin .NET attribute'lari kullanilmaktadir.

```
// [HttpGet] - GET isteklerini bu action karsilar
[HttpGet]
public IActionResult Login() => View();

// [HttpPost] - POST isteklerini bu action karsilar
[HttpPost]
public async Task<IActionResult> Login(LoginViewModel model) { ... }

// [Authorize] - login olmayan kullanicilari engeller
[Authorize]
public class FeedController : Controller { ... }

// [Authorize] - sadece bu action icin
[Authorize]
[HttpPost]
public IActionResult Create() { ... }

// [ValidateAntiForgeryToken] - CSRF saldirilarini onler
[HttpPost]
[ValidateAntiForgeryToken]
public async Task<IActionResult> JoinToggle(int id) { ... }
```

2. async / await (Tum Controller'larda)

Veritabani islemleri asenkron yapilmaktadir. 'async Task<IActionResult>' imzasi ve 'await' keyword'u her controller'da gecer.

```
// AuthController.cs
public async Task<IActionResult> Login(LoginViewModel model)
{
    // await: DB sorgusu tamamlanana kadar thread bloklamaz
    var user = await db.Users
        .FirstOrDefaultAsync(u => u.Email == model.Email);

    await SignInUser(user);    // cookie yazilana kadar bekle
    return RedirectToAction("Index", "Feed");
}

// ReviewsController.cs - transaction ile
public async Task<IActionResult> Create(int perfumeId, ...)
{
    using var tx = await db.Database.BeginTransactionAsync();
    db.Reviews.Add(new Review { ... });
    await db.SaveChangesAsync();
    await tx.CommitAsync();
    return RedirectToAction(...);
}

// PerfumesController.cs - LINQ + await
filter.Perfumes = await query
```

```
.Skip((filter.Page - 1) * filter.PageSize)
.Take(filter.PageSize)
.ToListAsync();
```

3. Action Donus Tipleri

```
return View(model);
// -> ViewModel'i ilgili .cshtml view'ina gönderir

return RedirectToAction("Index", "Feed");
// -> kullanicivi Feed/Index sayfasina yönlendirir

return PartialView("_PerfumeGrid", filter);
// -> AJAX isteklerinde sadece grid HTML'ini doner

return Json(results);
// -> parfum autocomplete endpoint'i JSON doner

return NotFound();    // 404
return Forbid();       // 403 - yetki yok
```

.NET Syntax Ornekleri (devam)

4. Dependency Injection - Primary Constructor

C# 12 primary constructor syntax'i ile servisler constructor parametresi olarak alinir, alan tanimina gerek kalmaz.

```
// Tek bagimlilik
public class AuthController(AppDbContext db) : Controller { }

// Birden fazla bagimlilik
public class CommunitiesController(
    AppDbContext db,
    ICurrentUserService currentUser
) : Controller { }

// Program.cs'de kayit
builder.Services.AddDbContext<AppDbContext>(...);
builder.Services.AddScoped<ICurrentUserService, CurrentUserService>();
builder.Services.AddScoped<INotificationService, NotificationService>();
```

5. ModelState - Sunucu Tarafi Validasyon

```
// Her POST action'inda validasyon kontrolu
[HttpPost]
public async Task<IActionResult> Register(RegisterViewModel model)
{
    if (!ModelState.IsValid) return View(model);
    //    ^--- [Required], [EmailAddress] vb. annotation
    //          kontrolleri burada gerceklesir

    // Ek is mantigi validasyonlari:
    if (await db.Users.AnyAsync(u => u.Email == model.Email))
        ModelState.AddModelError("Email", "Bu e-posta zaten kullanimda.");

    if (!ModelState.IsValid) return View(model);
    // ... kayit devam eder
}
```

6. EF Core LINQ Sorgulari

```
// Tek kayit getir
var user = await db.Users
    .FirstOrDefaultAsync(u => u.Email == model.Email);

// Kosullu liste
var reviews = await db.Reviews
    .Include(r => r.User)           // JOIN
    .Where(r => r.PerfumeId == id)  // WHERE
    .OrderByDescending(r => r.CreatedAt)
    .Take(10)
    .ToListAsync();

// Var mi kontrolu
bool exists = await db.Likes
```

```
        .AnyAsync(l => l.PerfumeId == id && l.UserId == userId);

// Sayım
int count = await db.Likes.CountAsync(l => l.PerfumeId == id);

// Gruplama
var brands = await db.Perfumes
    .GroupBy(p => p.Brand)
    .Select(g => new { Value = g.Key, Count = g.Count() })
    .ToListAsync();
```

Razor Syntax Ornekleri

7. @model - Tip Guvenligi

Her view'in en ustunde @model direktifi ile hangi ViewModel'i bekledigini belirtir. Bu sayede Model.Property erisimi tam tip guvencelidir.

```
<!-- Views/Perfumes/Details.cshtml -->
@model WebParfum.ViewModels.PerfumeDetailViewModel

<!-- View icinde kullanim -->
<h3>@Model.Perfume.Name</h3>
<p>@Model.Perfume.Brand</p>
<span>@Model.LikeCount begeni</span>
<span>@Model.Reviews.Count yorum</span>

<!-- Views/Communities/Details.cshtml -->
@model WebParfum.ViewModels.CommunityDetailViewModel

<h1>@Model.Community.Name</h1>
@if (Model.IsOwner) { <span>(sen)</span> }
@if (Model.IsMember) { <button>Ayril</button> }
```

8. @{ } - C# Kod Blogu

View icinde C# kodu calistirmak icin kullanilir.

```
<!-- Views/Perfumes/Details.cshtml -->
@{
    // Sayfa basligi dinamik olarak set ediliyor
    ViewData["Title"] = Model.Perfume.Name;

    // C# degiskeni tanimlama
    var placeholder = $"https://placeholder.co/400x400/f8e8d0/d97706"
        + $"?text={Uri.EscapeDataString(Model.Perfume.Brand)}";

    // Tuple array tanimlama
    var statusOptions = new[] {
        ("owned", "Sahibim", "bi-bag-check"),
        ("tried", "Denedim", "bi-droplet-half"),
        ("wishlist", "Istek Listem", "bi-bookmark-heart"),
    };
}
```

9. @if / @foreach / @for - Kontrol Yapilari

```
<!-- @if - kosullu render -->
@if (User.Identity?.IsAuthenticated == true)
{
    <button>Begeniyorum</button>
}

@if (Model.Perfume.Year.HasValue)
{
    <span>@Model.Perfume.Year</span>
}
```

```
}

<!-- @foreach - liste dongusu -->
@foreach (var r in Model.Reviews)
{
    <div>
        <strong>@r.User.Username</strong>
        <p>@r.Body</p>
    </div>
}

<!-- @for - sayici dongus (yildiz render) -->
@for (int i = 0; i < r.Rating; i++)
{
    <i class="bi bi-star-fill"></i>
}
}
```

Razor Syntax Ornekleri (devam)

10. Tag Helpers (asp-* attribute'lari)

Tag Helper'lar HTML etiketlerine asp-* attribute'lari ekleyerek URL uretimi ve form baglama islemlerini C# koduna baglar.

```
<!-- asp-controller + asp-action: URL otomatik uretilir -->
<a asp-controller="Perfumes" asp-action="Index">
    Kesfet
</a>
<!-- Cikti: <a href="/Perfumes">Kesfet</a> -->

<!-- asp-route-* ile parametre gecme -->
<a asp-controller="Perfumes" asp-action="Details" asp-route-id="@p.Id">
    @p.Name
</a>
<!-- Cikti: <a href="/Perfumes/Details/42">Sauvage</a> -->

<!-- Form - POST action'a gonderi -->
<form asp-controller="Reviews" asp-action="Create" method="post">
    ...
</form>
<!-- Cikti: <form action="/Reviews/Create" method="post"> -->

<!-- asp-for: input'u ViewModel property'sine bagla -->
<input asp-for="Email" class="form-control" />
<!-- Cikti: <input name="Email" id="Email" type="email" ... /> -->

<!-- asp-validation-for: hata mesaji alani -->
<span asp-validation-for="Email" class="text-danger"></span>
<!-- Validasyon hatasi varsa burada goster -->

<!-- asp-route-* ile accord filtre linki -->
<a asp-controller="Perfumes" asp-action="Index" asp-route-Accord="@accord">
    @accord
</a>
```

11. @Html Helper'lari

```
<!-- @Html.AntiForgeryToken() - CSRF koruma token'i -->
<form asp-action="Toggle" method="post">
    @Html.AntiForgeryToken()
    <!-- Cikti: <input type="hidden" name="__RequestVerificationToken" ... /> -->
    <button type="submit">Begin</button>
</form>

<!-- @Html.Raw() - HTML encode edilmeden render et -->
<div class="note-chips">
    @Html.Raw(RenderNoteChips(Model.Perfume.TopNotes))
</div>
<!-- C#'da olusturulan <span> HTML'i direkt basiliyor -->

<!-- @Html.PartialAsync() - baska bir view dosyasi dahil et -->
@await Html.PartialAsync("_ValidationScriptsPartial")
```



```
<!-- jQuery Validation script'lerini sayfaya ekler -->
```

12. @functions {} - View Icinde C# Metodu

```
<!-- Views/Perfumes/Details.cshtml -->
@functions {
    // View icinde kullanılacak yardimci metod
    string RenderNoteChips(string? raw)
    {
        if (string.IsNullOrEmpty(raw)) return string.Empty;
        var sb = new System.Text.StringBuilder();
        foreach (var n in NoteVisual.Split(raw))
        {
            var s = NoteVisual.Get(n);
            sb.Append($"<span class='note-chip' style='--note-color:{s.Hex}'>");
            sb.Append($"<i class='bi {s.Icon}'></i>");
            sb.Append($"<span>{n}</span></span>");
        }
        return sb.ToString();
    }
}

<!-- View'da kullanim -->
@Html.Raw(RenderNoteChips(Model.Perfume.TopNotes))
```

13. @section Scripts - Sayfa Bazli JS

```
<!-- Views/Auth/Register.cshtml -->
@section Scripts {
    <!-- jQuery Validation - [Required] vb. client'ta calisir -->
    @await Html.PartialAsync("_ValidationScriptsPartial")
}

<!-- Views/Perfumes/Index.cshtml -->
@section Scripts {
    <script>
        // Sadece bu sayfaya ozel JS
        const filterForm = document.getElementById('filterForm');
        // AJAX filtre, chip toggle, brand dropdown ...
    </script>
}

<!-- _Layout.cshtml'de bu section render edilir -->
@await RenderSectionAsync("Scripts", required: false)
<!-- required:false = section yoksa hata vermez -->
```

Syntax Ozet Tablosu

.NET Syntax (.cs dosyaları)

Syntax	Ne Yapar	Nerede
[HttpGet] / [HttpPost]	HTTP metod attribute'u	AuthController, PerfumesController
[Authorize]	Login zorunlulugu	Feed, Collection, Communities...
[ValidateAntiForgeryToken]	CSRF koruma	Tum POST action'lari
async Task<ActionResult>	Asenkron action imzasi	Tum controller action'lari
await ... Async()	Asenkron DB bekleme	FirstOrDefaultAsync,ToListAsync...
ModelState.IsValid	Validasyon kontrol noktası	Login, Register, Create action'lari
ModelState.AddModelError	Manuel hata ekleme	Duplicate email/username kontrolu
return View(model)	View'a ViewModel gonder	Tum controller'lar
return RedirectToAction	Sayfaya yonlendir	Basarili islem sonrası
return Forbid()	403 - Yetki yok	Gonderi silme yetki kontrolu
return Json()	JSON yanıt	Autocomplete endpoint
return PartialView()	Partial HTML - AJAX için	PerfumeGrid AJAX
Primary Constructor DI	Servis injection	Tum controller'lar
LINQ + EF Core	DB sorgu dili	Where, Include, GroupBy, Take...

Razor Syntax (.cshtml dosyaları)

Syntax	Ne Yapar	Nerede
@model ...	ViewModel tipi bildirimi	Tum view'larin 1. satiri
@{ }	C# kod blogu	ViewData, degisken tanimlama
@Model.Property	ViewModel property erisimi	Her yerde - @Model.Perfume.Name vb.
@if () { }	Kosullu HTML render	Auth kontrolu, null kontrol
@foreach () { }	Liste dongusu	Yorumlar, parfümler, uyeler
@for () { }	Sayici dongus	Yildiz puani render
asp-controller/action	URL uretme Tag Helper	Tum link ve form'larda
asp-for	Input-ViewModel baglama	Login, Register, Create formlari
asp-validation-for	Hata mesajı alani	Form validasyon span'lari
asp-route-*	URL parametresi ekleme	Details/id, Accord filtre linki
@Html.AntiForgeryToken	CSRF token	POST form'larında
@Html.Raw()	Ham HTML render	Nota chip HTML'i
@Html.PartialAsync()	Partial view dahil etme	ValidationScripts, _PerfumeGrid
@functions { }	View icinde C# metodu	RenderNoteChips metodu
@section Scripts { }	Sayfa bazli JS blogu	Her sayfanin kendi JS'i
@User.Identity...	Auth bilgisi okuma	IsAuthenticated, Name
ViewData["Title"]	Sayfa basligi set etme	Her view'in @{ } blogu